

Designing Personal Informatics in Context: Toward Collective Well-being through Nuanced Personal Data

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Personal informatics systems have the potential to support well-being by helping individuals better understand themselves. However, most current systems focus on quantifiable behavioral data, making it difficult to capture the implicit, contextual, and subjective aspects that underlie meaningful self-awareness. In my dissertation, I explore how self-tracking technologies can be designed to promote eudaimonic well-being through the collection and reflection of nuanced personal data embedded in users' lived experiences. I examine how personal informatics tools can be adapted to users' everyday contexts, interpersonal dynamics, and workplace environments through a series of empirical studies and system deployments. These studies investigate how individuals engage with their data, how social feedback expands self-understanding, and how organizational structures can support or constrain personal well-being efforts. This work contributes a multi-layered perspective on personal informatics that integrates intrapersonal, interpersonal, and institutional dimensions of well-being, and opens up new possibilities for designing sustainable and socially embedded self-tracking systems.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: Well-being, Personal Informatics, Nuanced Personal Data, Qualitative

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1 Context and Motivation

The concept of well-being encompasses optimal psychological functioning across various life domains essential for a fulfilling life [28]. Well-being is commonly viewed through two complementary perspectives: hedonic well-being, focused on happiness and satisfaction, and eudaimonic well-being, centered on self-realization, personal growth, and meaningful life pursuits [32]. Personal informatics (PI), defined as the practice of collecting, analyzing, and reflecting on personal data for meaningful insights, represents an emerging field closely aligned with the pursuit of well-being [10, 20]. PI activities enable individuals to extend their understanding of themselves and foster a more nuanced sense of self-awareness. While PI can potentially support both hedonic and eudaimonic dimensions depending on individual motivations [26], this paper positions personal informatics primarily as a meaningful approach to achieving eudaimonic well-being through nuanced personal data practices.

The scope of personal informatics has significantly expanded due to advancements in digital technology, enabling effortless data collection on behaviors, environments, and biometrics [5]. For instance, smartphone apps and wearable devices routinely gather quantifiable data on physical activity, sleep patterns, and heart rate, enhancing self-monitoring convenience and effectiveness [6, 7, 15]. Concurrent advances in artificial intelligence have simplified data analysis,

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providing personalized insights that substantially deepen self-understanding [8, 16]. As a result, these technological advancements enhance self-awareness through meaningful insights while reducing the effort required for sustained self-tracking, ultimately helping individuals move closer to achieving overall well-being.

Despite technological progress, achieving comprehensive self-understanding and well-being through solely quantitative methods remains challenging [4]. First, current technologies inadequately capture nuanced contextual data crucial for interpreting subjective experiences. For example, while wearable devices might detect physiological signals indicative of stress indicators, they often struggle to identify underlying contextual triggers [13, 14]—whether stemming from work obligations, family dynamics, or environmental factors. Second, individuals inherently possess “blind areas”—aspects known to others but unrecognized by themselves, such as non-conscious behaviors and communication styles, as described by the Johari Window model [21]. Addressing these blind spots requires external feedback, highlighting the limitations of relying solely on self-generated data. Lastly, even when individuals strive for enhanced self-awareness and improved well-being, they frequently encounter external constraints beyond their control, such as interpersonal conflicts at work or restrictive organizational cultures [2, 33]. Thus, for a truly comprehensive understanding of the self and the promotion of eudaimonic well-being, it is crucial to capture rich, qualitative, and contextually embedded personal data that reflects an individual’s lived experiences.

In my dissertation, I aim to design and explore personal informatics systems tailored to enhance well-being within individuals’ contextual realities across intrapersonal, interpersonal, and institutional domains. Additionally, I intend to investigate how personal efforts toward well-being might reciprocally influence social and organizational well-being. To achieve this, I propose three research questions:

- RQ 1. How should self-tracking tools be designed to capture implicit and subjective aspects of stress and psychological states encountered in everyday life?
- RQ 2. How can personal informatics tools involving external participation gather data that individuals struggle to self-generate or recognize independently? Furthermore, what impact does interpersonal involvement have on mutual social well-being?
- RQ 3. Within organizational environments resistant to substantial change, how can personal informatics systems be effectively integrated, and what role can organizations play in supporting individual well-being? Additionally, how does organizational support for individual well-being influence broader organizational wellness?

2 Capturing Implicit and Subjective Data for Individual Well-being (RQ1)

Mental health is as crucial as physical health for individual well-being, and stress significantly affects mental well-being [3, 19]. Accurately understanding daily behaviors associated with stress is vital yet challenging due to the invisible, ephemeral, and subjective nature of mental health data [18]. To address this, I have designed self-tracking tools for capturing and managing mental health data within everyday contexts, such as academic life and workplaces.

In Study 1, recognizing the link between daily stressors and scheduled activities, we developed a calendar-based stress anticipation application [17]. This tool enables users to anticipate stressful events, prepare intervention plans, and compare actual stress levels against expectations. A 30-day deployment with 47 university students revealed that surfacing stressors proactively aided participants in identifying triggers and significantly reduced stress through the planning and executing of personalized interventions. This work contributes to understanding the potential role of prospective and future-centric approaches within personal informatics systems.

Study 2 (ongoing) focuses on a mobile app tailored for call center employees who frequently face emotional dissonance due to expressing mandated emotions while suppressing genuine ones [9, 11]. An extensive formative study using

surveys and focus group interviews informed the design requirement of the application, featuring easy emotional state logging and evidence-based microbreak interventions (within 2 minutes each). Preliminary findings from a 6-week deployment with 46 call center workers indicated improved self-awareness regarding authentic emotions and contextual triggers. The tracking process itself served as a healing activity, providing brief yet meaningful emotional relief. Regular usage of the application also facilitated workers in developing personalized coping strategies, significantly reducing depression in a high-stress environment.

3 Extending Self-Awareness Through Reciprocal Social Tracking (RQ2)

Through my previous studies, I confirmed that self-tracking nuanced personal data significantly enhances self-understanding and promotes well-being. However, a critical question remains: Can self-tracking alone lead to complete self-understanding? Individuals inherently possess “blind areas”—aspects visible to others but unnoticed by themselves—suggesting the necessity of external perspectives. Personal Informatics (PI) research has explored how social features help self-trackers gain new insights into their personal data by sharing with social networks (e.g., friends, family, or online communities) [22, 24, 25, 30]. Additionally, caregivers often facilitate tracking for children or patients who find self-tracking challenging [12, 23, 29]. However, while assistance from caregivers might be considered natural or expected, consistently involving others in the everyday tracking process remains difficult.

To address this, Study 3 (forthcoming in CHI25) proposes reciprocal tracking, a mutual exchange of meaningful observations and insights between pairs of participants closely familiar with each other’s contexts. In a week-long study with 9 dyads (18 participants), reciprocal tracking provided participants with contextualized insights into their behaviors, expanding their self-perception by revealing previously unnoticed behaviors, clarifying vague self-understandings, and enhancing awareness of how others perceive their actions. The study also highlighted the impact of reciprocal tracking on interpersonal relationships and social well-being. Positive outcomes included increased mutual interest, shared activities, and stronger emotional bonds through deeper conversations. Conversely, notable challenges arose, illustrated by a case where one pair withdrew due to feedback that inadvertently caused emotional hurt and reduced mutual trust. Thus, this work contributes to personal informatics by introducing reciprocal tracking as an approach that ensures mutual benefits and broadens the scope of social features within personal informatics systems.

4 Future Work (RQ3)

Through Study 2, I found that call center employees improved their intrapersonal well-being through emotion management. However, despite personal efforts, substantial improvements were limited by external factors beyond individual control, including heavy workloads, interactions with aggressive customers, and workplace conflicts [1, 34]. This raises the question: How can personal informatics systems support individual well-being in such challenging contexts?

Employee well-being is closely connected to organizational health. Improved employee well-being fosters positive work attitudes and increased productivity, benefiting overall organizational performance and culture [27, 31]. Yet, organizations typically lack effective methods to understand factors influencing employees’ well-being beyond performance metrics, often relying on gut feelings based on workers’ facial expressions or performance declines.

To address this, I will collaborate with a call center employing over 40 workers, deploying the emotional state tracking tool from Study 2 for at least 12 weeks. By analyzing aggregated data, I will provide narrative reports detailing emotional distributions, influencing factors, and effective micro-intervention activities, ensuring privacy by sharing only collective data. This approach aims to facilitate organizational understanding of employees’ emotional states, encouraging the design and implementation of targeted interventions at departmental or center levels, and contributing

to long-term organizational strategies. Expected outcomes include gaining contextual insights previously unavailable to organizations, identifying effective data-driven micro-interventions, evaluating their impact on employee well-being, and exploring how organizational support for individual well-being enhances overall organizational health.

5 Expected Contribution

This dissertation contributes to the advancement of personal informatics by reimagining how self-tracking technologies can support well-being through richer, more contextually grounded understandings of the self. By focusing on nuanced and subjective aspects of human experience, it addresses the limitations of conventional systems that primarily rely on quantitative data. Through a series of empirical studies and system designs, this work demonstrates how self-awareness can be deepened through interpersonal feedback and how organizations can play a constructive role in supporting individual well-being. Overall, the research offers a comprehensive, multi-layered perspective on personal informatics that spans intrapersonal, interpersonal, and institutional domains.

In particular, this work acknowledges the burden often associated with self-reporting and explores ways to reduce it by designing data collection experiences that align with users' natural behaviors, everyday tools, and contextual environments. Rather than imposing rigid input structures, the systems developed in this dissertation aim to support more fluid and sustainable engagement through user-centered data collection strategies. These design directions open important questions around how we might further balance manual and automated methods, personalize data capture experiences, and support meaningful yet low-effort reflection in everyday life [4]. I look forward to discussing these challenges—and their broader implications—through the Doctoral Consortium.

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